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Research Paper

Nesting Mottled Duck (*Anas fulvigula*) selection of ungrazed grasslands in southwestern Louisiana is associated with increased nest survival

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ABSTRACT. Nest site selection is a discrete and often repeated choice, and individuals should select nest sites that maximize reproductive success and thus increase fitness. Mottled Ducks (*Anas fulvigula*) are a non-migratory species that inhabits the Gulf Coast of the United States year-round and therefore have the ability to constantly evaluate habitat to make well-informed nest site choices compared to migratory species. Mottled Duck populations have declined over the last decade and a better understanding of nest site selection and its relationship to nest survival is a top research priority. We deployed GPS transmitters on 148 females across three breeding seasons to evaluate nest site habitat selection and nest survival. We observed 30 nest attempts and found females selected sites in diverse landscapes, but Mottled Ducks preferred old fields and pasture relative to other habitats. High vegetation density surrounding the nest bowl had a positive influence on nest survival. We found that females were more likely to renest when the initial nest failure occurred earlier in the incubation period. Our results emphasize the importance of preserving tall, dense vegetation in upland habitats. Additionally we recommend that prescribed burns are timed to ensure adequate vegetative cover for Mottled Ducks by the onset of nesting in March.

Sélection par le Canard brun (*Anas fulvigula*) de prairies non broutées dans le sud-ouest de la Louisiane associée à une augmentation de la survie des nids

RÉSUMÉ. Le choix du site de nidification est une action distincte et souvent répétée, et un individu devrait choisir un site de nidification qui maximise son succès de reproduction et, du même coup, augmente son aptitude phénotypique (fitness). Le Canard brun (*Anas fulvigula*) est une espèce non migratrice qui habite la côte États-Unienne du golfe du Mexique toute l'année et qui a donc la capacité d'évaluer constamment l'habitat pour choisir avec attention un site de nidification contrairement aux espèces migratrices. Les populations de Canards bruns ont diminué au cours de la dernière décennie et une meilleure compréhension de la sélection des sites de nidification et de sa relation avec la survie des nids sont prioritaires. Nous avons muni d'émetteurs GPS 148 femelles au cours de trois saisons de nidification afin d'évaluer la sélection de l'habitat du site de nidification et la survie du nid. Nous avons observé 30 tentatives de nidification et constaté que les femelles choisissaient des sites variés, mais qu'elles préféraient les vieux champs et pâturages. Une forte densité de végétation autour du nid a eu un effet positif sur sa survie. Nous avons remarqué que les femelles étaient plus susceptibles de nicher une seconde fois lorsque l'échec du premier nid se produisait tôt dans la période d'incubation. Nos résultats soulignent l'importance de préserver une végétation haute et dense dans les milieux secs. En outre, nous recommandons que les brûlages dirigés soient programmés de manière à assurer une couverture végétale adéquate pour le Canard brun avant le début de la nidification en mars.

Key Words: *breeding ecology; coastal marsh; Gulf Coast; habitat; waterfowl*

INTRODUCTION

Avian nest site selection is a discrete choice that has direct implications for adult survival and reproductive output (Martin 1993). Accordingly, nest site preferences have been shaped by natural selection and birds have evolved numerous strategies for assessing and mitigating the risk of predation (reviewed in Lima 2009). Evaluating the adaptive significance of nest site selection requires information on the suite of options available to a nesting bird, and the fitness outcome of choosing a particular site (Clark

and Shutler 1999). This can be accomplished by comparing habitat characteristics of successful and unsuccessful nests, and simultaneously diagnosing preference by comparing characteristics of used versus available habitat (Cody 1985, Chalfoun and Schmidt 2012).

Predation is the most common source of nest failure in North American ducks (Klett et al. 1988, Greenwood et al. 1995) and is a primary driver of population dynamics (Hoekman et al. 2002) so nest site choices are under strong selective pressure. Nest site

selection is hierarchical (Eicholz and Elmberg 2014), and although patterns of dabbling duck nest survival are well-known across spatial scales (Ringelman et al. 2018), the extent to which ducks can or do make adaptive decisions remains an open question (Clark and Shutler 1999, Devries et al. 2018), especially given spatio-temporal variation in predation risk (Walker et al. 2013, Ringelman et al. 2017). Nest site preferences may be innate, can be modified by current cues about predation risk (Eicholz et al. 2012), and may change based on public information cues (Pöysä 2006, Ringelman et al. 2014) or prior experience (Lokemoen et al. 1990, Majewski and Beszterda 1990, Ringelman et al. 2017).

In the Prairie Pothole Region, ground-nesting dabbling ducks have higher nest survival in landscapes with a large proportion of intact grassland (Stephens et al. 2005, Ringelman et al. 2018) with few edges (Pasitschniak-Arts et al. 1998, Stephens et al. 2004) because predators concentrate their foraging efforts in other habitats, such as isolated patches of cover or wetland-agricultural edges, where they are more efficient (Phillips et al. 2003, 2004). At the site level, nest survival is lower in areas that provide denning habitat (Larivière and Messier 1998a, b) or travel corridors (Larivière and Messier 2000, Raquet et al. 2015, Ludlow and Davis 2018) for predators. Nest site characteristics can affect nest survival (Crabtree et al. 1989, Clark and Shutler 1999, Setash et al. 2020), but the influence of factors like vegetation height or density is remarkably inconsistent among studies (reviewed in Ringelman and Skaggs 2019), despite the fact that it should affect visual (Borgo and Conover 2016) and olfactory (Nams 1997) detection by predators. Finally, the adaptive significance of a nest site choice depends on the decisions of others, because local nest density may affect nest survival in the context of the predator community (Ringelman et al. 2012, Ringelman 2014) and time of year (Ringelman et al. 2018).

The evidence that ducks make adaptive nest site choices is inconsistent (Clark and Shutler 1999, Borgo and Conover 2016, Devries et al. 2018, Dyson 2020, Setash et al. 2020), and this may be in part because most ducks are long-distance migrants and have relatively little time to evaluate a temporally unpredictable landscape and select nest sites. In contrast, Mottled Ducks (*Anas fulvigula*) are one of the few non-migratory ground-nesting ducks in North America, found along the Gulf Coast of the southern United States (Baldassarre 2014). Because Mottled Ducks have the potential to evaluate habitats over longer periods to make adaptive choices, Mottled Ducks should theoretically make well-informed nest site selection decisions. However, Mottled Duck nest success is highly variable among habitats and years (reviewed in Bonczek and Ringelman 2021), ranging from 5% in southwestern Louisiana (Baker 1983) to 31% in the Atchafalaya River Delta (Holbrook et al. 2000). Mottled Ducks must contend with a diversity of nest predators, including raccoons (*Procyon lotor*), coyotes (*Canis latrans*), striped skunks (*Mephitis mephitis*), snakes, and American Alligators (*Alligator mississippiensis*; Walters et al. 2001, Finger et al. 2003, Dugger et al. 2010, Shipes et al. 2015, Kneece 2016). Other causes of nest loss include flooding (Caillouet 2015), mowing/plowing operations (Dugger et al. 2010), and trampling by cattle (Durham and Afton 2003).

Previous studies on Mottled Duck nesting ecology have documented nests in a variety of landscape types, including prairie, idle rice fields, cattle pasture, and marsh in Texas and

southwestern Louisiana (Stutzenbaker 1988, Finger et al. 2003, Bonczek and Ringelman 2019), and on dredge spoil islands, canal banks, and levees in the river deltas of Louisiana (Holbrook et al. 2000, Walters et al. 2001). In southwestern Louisiana, female Mottled Ducks frequently nest in idle fields or permanent pastures with knolls, and select sites with taller vegetation (Durham and Afton 2003). On the mid-coast of Texas, nests were also found in vegetation > 60 cm tall with > 75% canopy cover, with resultant high apparent nest success of 32% (Finger et al. 2003). Mottled Ducks nesting in marsh or other areas like fallow rice fields that are prone to flooding may elevate their nest bowl (Finger et al. 2003, Bonczek and Ringelman 2019), but fail to do so on dredge-spoil islands (Caillouet 2015).

The Western Gulf Coast Mottled Duck population has declined by ~50% since breeding surveys began in 2008, and a better understanding of nest site selection and its relationship to nest survival is a top research priority laid out in the Mottled Duck Conservation Plan produced by the Gulf Coast Joint Venture (Wilson 2007). Understanding the linkage between landscape characteristics, habitat conditions, and vital rates is a critical step in designing effective conservation and management plans for Mottled Ducks. Although previous research has examined landscape characteristics that influence nest survival, very few studies (Holbrook et al. 2000, Walters et al. 2001, Durham and Afton 2003) have investigated the factors that influence nest site selection in Mottled Ducks. Additionally, previous studies have located Mottled Duck nests by searching targeted habitats or observing drop flights as females return to their nests after morning incubation breaks. These techniques therefore limit our understanding of which landscapes Mottled Ducks truly prefer across the region, because ducks may nest in landscapes not under observation. To address these shortcomings, we used GPS-GSM transmitters as a novel means of monitoring nesting Mottled Ducks in southwest Louisiana, which is viewed as a stronghold for the Western Gulf Coast population. The objective of our study was to determine how local and landscape-level factors influence nest site selection and nest survival, and to characterize re-nesting behavior following nest failure.

METHODS

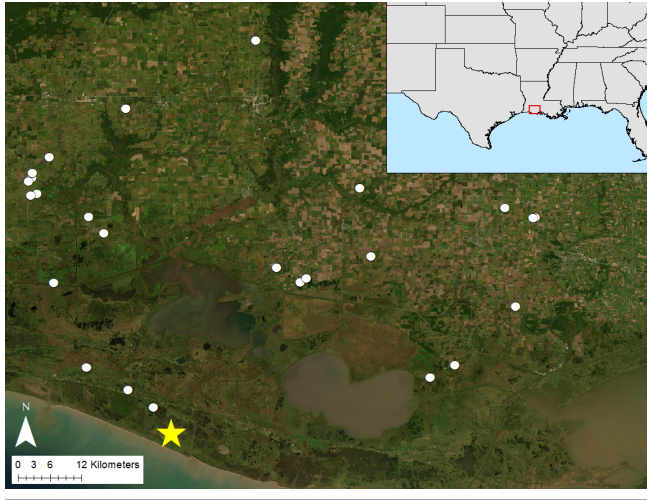
Study Site

Our primary Mottled Duck capture site was at Rockefeller Wildlife Refuge (hereafter Rockefeller) in southwestern Louisiana, a roughly 29,000 ha property managed by the Louisiana Department of Wildlife and Fisheries. Rockefeller is located in Cameron and Vermilion Parishes at the southern end of the Mermentau River Basin near Grand Chenier, LA; Rockefeller borders the Gulf of Mexico and extends inland ~10 km. Rockefeller contains both managed and unmanaged wetland impoundments that vary from fresh to saline, and moist soil management units (Selman et al. 2011). The birds we marked nested across southwestern Louisiana in Cameron, Vermilion, Calcasieu, and Jefferson Davis parishes (13,528 km²; Fig. 1).

Data collection

In conjunction with annual waterfowl banding activities conducted by Louisiana Department of Wildlife and Fisheries, we captured adult female Mottled Ducks on Rockefeller and surrounding private land July–September of 2017–2019 from

Fig. 1. White circles indicate the location of nesting attempts by female Mottled Ducks (*Anas fulvigula*) monitored in southwestern Louisiana, USA, 2018–2021. Rockefeller Wildlife Refuge, where most ducks were captured during molt, is marked with a star.



airboats using night-lighting techniques while birds were molting and flightless (Cummings and Hewitt 1964). We banded captured individuals with a U.S. Geological Survey aluminum or incoloy leg band and some individuals received a 21 g solar-powered Saker-L GPS-GSM transmitter (Ecotone Telemetry, Gdynia, Poland), which uses the cellular network to transmit locations and were accurate to 30 m. We deployed transmitters on birds ≥ 690 g to ensure units were $\leq 3\%$ of body mass and prioritized deploying transmitters on Mottled Ducks that were nearly or fully molted. We attached the transmitters in a modified Dwyer-style configuration (Dwyer 1972) with two separate loops—one around the body behind the wings and one around the neck that sat at the base of the furcula; these loops were secured with knots and polyacrylamide glue. We used Conrad-Jarvis 6 mm black nylon automotive elastic with neoprene elastomer (Casazza et al. 2020) and tightened until the tip of the thumb fit under the unit. We released marked birds within 12 hours of capture at the original capture site. Transmitters logged global positioning system (GPS) coordinates every two hours during daylight and we obtained transmitted data from the manufacturer-designed web-based Ecotone data management panel. From this panel, we could obtain transmitter metrics, such as battery level, activity level, temperature, and date of most recent GPS location, and download individual .kmz files with monthly GPS locations viewable in Google Earth or .csv files of bird locations for further analysis. Our procedures were approved by Louisiana State University Institutional Animal Care and Use Committee under permit A2016-27 and federal bird banding permit 06669.

Monitoring began 1 Feb, by which time individuals had a minimum of five months of acclimation to the transmitter, and continued until 31 July 2018–2021. We examined the locations and movements of each individual every three days using the Google Earth .kmz file. Once we identified an individual that logged locations in the same area consecutively, we waited 10 days

from the inferred date of nest initiation, which was determined by the first date the female was observed at the presumed nest site, to ensure the female had begun incubating and then visited the site on foot to confirm the presence of a nest. Some nests were difficult to find and we used a trained dog to help find nests (Glover 1956, Keith 1961) if we were unable to locate them; these were typically nests with an absent female or nests that had already failed at the time of the first visit. At the initial nest visit we determined clutch size and incubation stage (in days) by candling (Weller 1956). For nests that had already been depredated, we recorded the number of depredated eggs present to obtain a minimum clutch size and use in conjunction with the telemetry data to infer initiation date. We also recorded the landscape type, which we defined as pasture (heavily grazed field), old field (lightly grazed or ungrazed field), emergent marsh, idle rice (moist soil plants), or crawfish (*Procambarus* spp.) pond (freshwater managed impoundment with emergent vegetation) levee, the presence or absence of cows, and if the nest was located overwater. Landscape types were designated in the field by a combination of land use and shared characteristics such as plant species and presence of water. Previous research demonstrated Mottled Ducks select habitat associated with recent cattle grazing (Haukos et al. 2010) and cattle stocking rates may influence nest survival (Durham and Afton 2003). At the nest site, we measured visual obstruction in decimeters (dm) as an index of vegetation density using a Robel pole at a height of 1 m and a distance of 4 m from the nest bowl in each cardinal direction (Robel et al. 1970). We recorded height of the tallest vegetation adjacent to the nest bowl and the top three dominant plant species and their percent cover within a 4 m radius around the nest bowl. We also determined elevation using a Hiper II dual-frequency Real Time Kinematic and static GNSS receiver from Topcon systems, used in static mode on a 1.6 m range pole with support bipod. The receiver was set to log all available GNSS satellite signals at 1-second intervals to an internal microSD card, and each static occupation was a minimum of 15 minutes. We were unable to determine elevation in 2020 because of complications obtaining the equipment related to the COVID-19 pandemic.

We monitored active nests using the GPS locations of the nesting female, and once the female no longer logged locations at the nest site, we returned to the nest to check the fate and collected the same measurements we gathered at the time of discovery. For nests that were depredated when found, we estimated the initiation date as the first day that we detected the female at the nest site using the transmitted GPS locations. We determined the exact fate day by the last day that we observed the female at the nest site from the GPS locations. From satellite imagery and using the Euclidean Distance tool in ArcMap version 10.8.2 (Esri, Redlands, CA, USA), we measured patch size of dominant habitat, distance to water, type of nearest water, distance to main road (defined as public access roads), distance to road (including two-tracks and driveways), distance to path (e.g., patch edge, levee, etc.), and distance to nearest croplands as defined by the 2019 National Land Cover Database (Dewitz and U.S. Geological Survey 2021). We used transitions between cover types to designate patch borders and thus calculate patch size. All means are reported with standard deviation and beta estimates with standard error unless otherwise specified.

Nest site selection

For every nest site, we selected four matched random points and visited them as close to the date of the initial nest site visit as possible; however, because of landowner permission and private land access, we were not able to obtain four random points for all nests. We selected these random points using Program R version 3.4.3 (R Core Team 2022) by generating an 80% minimum convex polygon home range of the nesting female for the six days prior to nest initiation. We used a six day window because this is the timeframe in which rapid follicular growth occurs in Mottled Ducks (Alisauskas and Ankney 1992), so we assumed females would be prospecting for a nest site during this time. Using the home range of the female allowed us to examine nest site selection at a biologically relevant scale. We took identical measurements at the random points as we did at the nest site. The landscape type of random points were classified in the same way as nests, in addition to an “other” category that encompassed open water, cropland, and forest.

To derive estimates of relative probability of use, we used generalized linear models with a log-link to check for differences between the nest sites and random points, where “1” and “0” designated used and available, respectively. We defined salinity based on a marsh vegetation type classification layer from the U. S. Geological Survey (Enwright et al. 2015), which delineated coastal marsh vegetation communities into saline, brackish, intermediate, freshwater marsh, and other. Areas classified as “other” we assumed to be fresh because all were located inland, and those classified as open water, which were not associated with a salinity within the marsh vegetation type layer, we determined to be the salinity of the nearest designation. Visual obstruction and vegetation height were correlated as were distance to main road, road, path, and patch size (all $r > 0.70$). Therefore, we included only visual obstruction and patch size in our models based on preliminary analyses of model fit and for comparison with previous studies. We included the quadratic term of visual obstruction because previous studies found that nesting birds may select an intermediate level of vegetation height and density (Götmark et al. 1995, Clark and Shutler 1999). We created a candidate set of nine univariate models and included models with multiple covariates when we determined the combination of covariates was biologically meaningful. We used Akaike information criterion corrected for small sample size (AICc) to rank models (Burnham and Anderson 2002), where models $\leq 2 \Delta\text{AICc}$ of the top model were considered competitive.

Nest survival

We modeled nest survival using the RMark package in Program R version 4.1.2 (White and Burnham 1999, Dinsmore et al. 2002, Laake 2013, R Core Team 2022) to estimate the daily survival of Mottled Duck nests. We constructed univariate models using a combination of local variables including average visual obstruction, dominant plant species, and whether the nest was located over water (wet). Landscape-level covariates included landscape type, distance to road, distance to water, patch size, and presence of cows (cows); other variables included initiation date and year (as a holistic proxy for environmental conditions). We constructed additional models including multiple covariates when we expected the combination of variables to be biologically

meaningful. We used visual obstruction readings collected at nest discovery given the inherent bias that the vegetation at hatched nests grows for longer, and is therefore higher/denser than at failed nests (Gibson et al. 2016, Ringelman and Skaggs 2019). We also added the quadratic term for visual obstruction because previous research found that nesting ducks may adaptively select an intermediate level of vegetation height and density (Clark and Shutler 1999). We ranked the resulting models using AICc score and considered models $\leq 2 \Delta\text{AICc}$ to be competitive.

Renesting propensity

We estimated renesting propensity as the proportion of females that were unsuccessful on their first nest attempt that subsequently initiated a second nesting attempt. We used linear regression to model the effects of initiation date of the first nest, incubation stage at failure of the first nest, and clutch size of the first nest and ranked the resulting models using AICc scores. We calculated the renesting interval as the time between the failure of the first nest and initiation of the second nest and reported the average interval with standard deviation. We were not able to model differences among years because there were no renesting attempts in 2018. We used a Welch two sample t-test to examine differences in clutch size between first and second attempt nests.

RESULTS

We deployed transmitters on 69, 58, and 21 new individuals in 2017, 2018, and 2019, respectively, and monitored 34, 37, 24, and 1 individuals over the course of the subsequent breeding season in 2018, 2019, 2020, and 2021, respectively. We located five nests that were initiated 13 March–5 June 2018 in addition to a dead female with an egg, 14 nests that were initiated 7 March–30 May 2019, nine nests that were initiated 30 March–5 July 2020, and one nest initiated on 7 April 2021 totaling 30 nest attempts during the entirety of the study (Fig. 1). We documented nest attempts over multiple years for four females. Three individuals showed strong site fidelity and nested in close proximity (23, 39, and 106 m) to the nest site of the previous year, two of which had previously hatched a nest at that site and the nest of the third individual hatched the second year under observation. The remaining female nested 7.2 and 19.4 km from previous first and second nest sites, respectively, and was not successful.

Eight of the 30 nests were already depredated upon initial location on foot. Average clutch size for incubated nests was 8.75 ± 1.34 eggs. Average number of eggshells located at nests that were depredated upon initial discovery was 5.83 ± 1.72 eggshells.

Nest site selection

Of the 30 nest attempts identified from telemetry data and confirmed on foot, we excluded two nests from analysis because of the inability to locate the nest bowl: one attempt was a dead female found with one egg but no visible nest bowl, and the other we were only able to locate eggshell fragments among extremely tall and dense vegetation. Mottled Ducks nested across southwestern Louisiana in various landscape types. Of the 28 nest sites we located, most nests were found in old fields (39.3%), followed by pasture (28.6%), emergent marsh (14.3%), idle rice (10.7%), and crawfish pond levees (7.1%; Table 1).

Table 1. Average visual obstruction measurement (dm) and common plant species by landscape type at the nest sites of Mottled Ducks (*Anas fulvigula*) in southwestern Louisiana, USA, 2017–2021.

Landscape type	Average visual obstruction (dm)	Common plant species
Old fields	6.3	<i>Andropogon</i> spp., <i>Paspalum</i> spp., <i>Eleocharis</i> spp.
Pasture	4.8	<i>Paspalum</i> spp., <i>Axonopus fissifolius</i> , <i>Spartina patens</i> , <i>Eleocharis</i> spp.
Emergent marsh	10.9	<i>Zizaniopsis milliacea</i> , <i>Sagittaria</i> spp.
Idle rice	4.3	<i>Eleocharis</i> spp., <i>Iva</i> spp.
Crawfish pond	3.5	<i>Panicum</i> spp., <i>Cynodon dactylon</i>

Elevations at nest sites ranged -0.07 – 6.22 m [$\bar{x} = 2.22 \pm 2.07$ m] and patch size ranged 0.6 – 2831 ha, [$\bar{x} = 515 \pm 1,067$ ha]. We removed an additional three nests from the nest site selection analysis because land access restrictions due to the COVID-19 pandemic in 2020 prevented us from obtaining random points, resulting in a total of 25 nests included in this analysis. Nest site selection in Mottled Ducks was best explained by landscape type and visual obstruction measurement, and no other models were competitive (Table 2). The probability of use was highest in old fields ($\beta = 2.452 \pm 1.462$) and lowest in emergent marsh ($\beta = -4.133 \pm 1.764$; Fig. 2). The probability of selecting a particular microhabitat was positively associated with visual obstruction at the nest bowl ($\beta = 0.658 \pm 0.178$; Fig. 3).

Table 2. Model results that best explained nest site selection of female Mottled Ducks (*Anas fulvigula*) in southwestern Louisiana, USA, 2017–2021. For simplicity, we only show the top model, the next best model, univariate models of the top-ranked model, and null model for comparison. Parameters in competitive models included visual obstruction measurement (robel), landscape type, and distance to nearest water.

Model	AIC _c	Δ AIC _c	AIC weight	K
Robel + landscape type	78.9	0.00	1	7
Robel + distance water	95.0	16.05	0	3
Landscape type	101.1	22.20	0	6
Robel	109.9	30.99	0	2
Null	123.9	44.99	0	1

Nest survival

The cause of failure for all nests was predation of the eggs ($n = 20$) or nesting female ($n = 2$), except for one nest that failed because of researcher-related abandonment. Over the course of the study, we monitored seven nests that hatched successfully. For our nest survival analysis, we excluded one nest because of researcher-related abandonment, two because we were unable to locate the nest bowl, and one because we only monitored a single nest during the 2021 season that was successful. Of the 26 nests included in the survival analysis, we identified five that failed during laying and 15 that failed during incubation. The top-ranked model of nest survival only included visual obstruction recorded at the nest (Table 3).

Fig. 2. Relationship between the probability of use and landscape type for Mottled Ducks (*Anas fulvigula*) nesting in southwestern Louisiana, USA, 2017–2021. Error bars denote standard error.

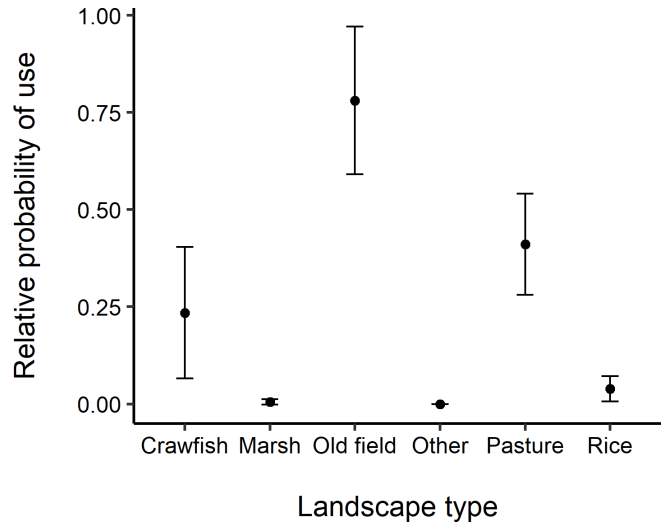
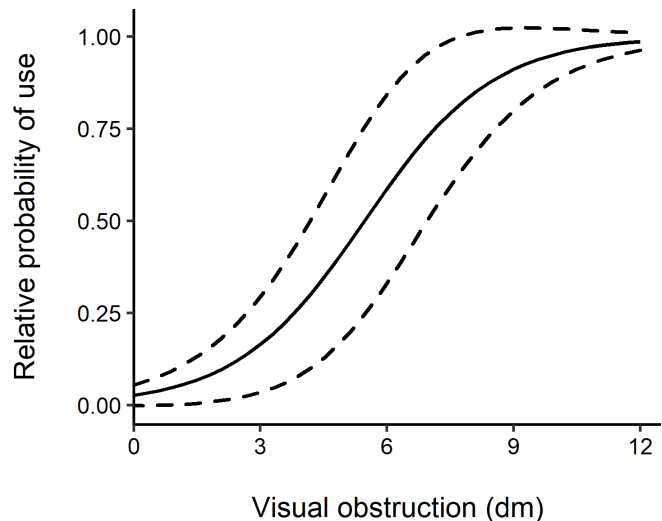


Fig. 3. Relationship between the probability of use and visual obstruction measurement (dm) for Mottled Ducks (*Anas fulvigula*) nesting in southwestern Louisiana, USA, 2017–2021. The dotted line represents the 95% confidence interval.

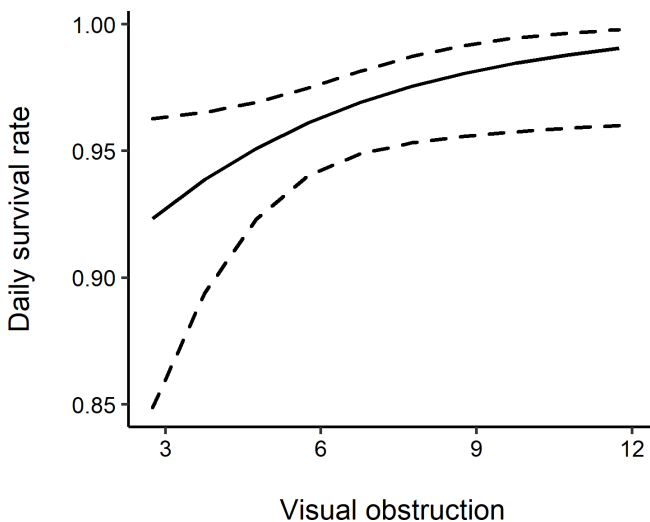


There were several models within 2Δ AIC_c that included visual obstruction in combination with some other variable (Table 3); however, the additional parameters did not improve model fit and beta confidence intervals overlapped zero so we judged these parameters to be uninformative (Arnold 2010). Our top model indicated that daily survival rate increased as visual obstruction increased ($\beta = 0.240 \pm 0.114$; Fig. 4).

Table 3. Model results within 2 ΔAIC_c , that best explain variation in daily survival of Mottled Duck (*Anas fulvigula*) nests in southwestern Louisiana, USA, 2018–2020. Parameters in competitive models included visual obstruction measurement (Robel), distance to cropland, patch size, whether the nest was located over water (wet), distance to water, presence or absence of cows (cows), and date of initiation (date).

Model	AIC _c	ΔAIC_c	AIC _c weight	K
Robel	164.4	0.00	0.193	2
Robel + distance crops	165.8	1.39	0.096	3
Robel + patch size	166.0	1.86	0.076	3
Wet	166.3	1.89	0.075	2
Wet + robel	166.4	1.97	0.072	3
Robel + distance water	166.4	1.98	0.072	3
Robel + cows	166.4	1.98	0.072	3
Robel + date	166.4	2.00	0.071	3
Null	168.0	3.60	0.032	1

Fig. 4. Relationship between visual obstruction measurements (dm) and daily survival rate of Mottled Duck (*Anas fulvigula*) nests in southwestern Louisiana, USA, 2018–2020.



The daily survival rate for nests in this study taken from our top-ranked model was 0.965 ± 0.008 , resulting in an overall nest success estimate of 0.284 for the 35-day nesting period. The only other notable model in our competitive set was the model “wet,” which differentiated between nests constructed overwater and those in upland habitats. The confidence interval overlapped zero, but the daily survival estimates for nests that were constructed overwater differed from those that were in upland habitat. The daily survival rate for overwater nests was 0.991 ± 0.009 , resulting in an overall nest success estimate of 0.729, while the daily survival rate for upland nests was 0.957 ± 0.010 , resulting in an overall nest success estimate of 0.215.

Renesting propensity

We documented seven individuals that initiated renesting attempts across all years, resulting in eight second-attempt nests (Fig. 5). Partitioned among years, we recorded zero, five, and three

Fig. 5. Nesting chronology for first and second nest attempts by Mottled Ducks (*Anas fulvigula*) in southwestern Louisiana, USA, 2018–2020. Mean first attempt nest initiation was on 15 April and second attempt nest initiation was on 28 May.

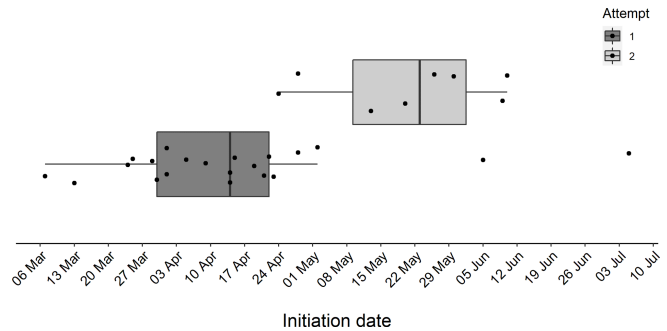


Table 4. Model results that best explain variation in renesting propensity of Mottled Ducks (*Anas fulvigula*) in southwestern Louisiana, USA, 2018–2020.

Model	AIC _c	ΔAIC_c	AIC _c weight	K
Incubation stage at failure	19.8	0.00	0.500	2
Null	21.5	1.66	0.218	1
Initiation date	22.8	3.03	0.110	2
Incubation stage at failure + initiation date	22.8	3.04	0.109	3
Clutch size	24.0	4.17	0.062	2

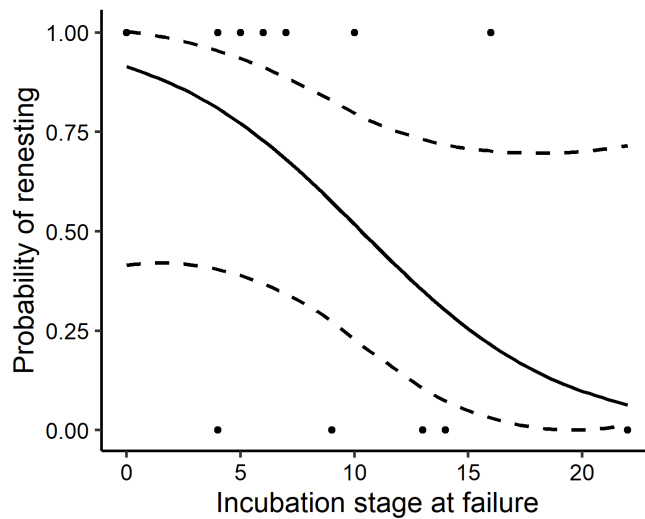
renest attempts in 2018, 2019, and 2020, respectively, resulting in renesting rates for unsuccessful females of 0.0, 0.71, and 0.75. The probability of renesting was best explained by how far into incubation the female was when the initial nest failed (Table 4). The null model was also within 2 AIC_c , however, the top model containing incubation stage at failure held more than twice the weight.

Females that were at a later stage of incubation at the time of nest failure were less likely to renest ($\beta = -0.2287 \pm 0.1323$; Fig. 6). The renesting interval between failure of the first nest and initiation of the second nest ranged from 14 to 38 days, averaging 26.0 ± 8.3 days. We found no association between the age of the first nest at the time of failure and renesting interval ($p = 0.76$) and no difference in the average clutch size of first and second nests ($p = 0.58$).

DISCUSSION

Nest site preferences in ducks are under strong selective pressure to convey advantages in survival and reproductive output, but preferences are only adaptive if ducks can sample habitats and evaluate risk with reasonable accuracy. We hypothesized that although Mottled Ducks may face a greater diversity of risk (from flooding, alligators) than their ground-nesting congeners breeding in the Prairie Pothole Region, as year-round residents of southwestern Louisiana, the Mottled Ducks we studied are uniquely positioned to evaluate that risk and make well-informed nest site selection decisions. Although there have been previous efforts to study habitat selection by Mottled Ducks, researchers searched for nests at specific sites, and primarily focused on

Fig. 6. Relationship between the probability of renesting and the incubation stage at which first nests failed for Mottled Ducks (*Anas fulvigula*) in southwestern Louisiana, USA, 2018–2020. The dashed line represents the 95% confidence interval.



selection of within-site habitat characteristics (Holbrook et al. 2000, Walters et al. 2001, Durham and Afton 2003), which precludes studying habitat selection at landscape-level scales, and could entirely overlook important landscape types used by Mottled Ducks for nesting.

Using a telemetry-based approach, we found that Mottled Ducks in southwest Louisiana nested in various landscape types, including the novel observation that some nests were constructed overwater in emergent marsh (Bonczek and Ringelman 2019). We found that Mottled Ducks were most likely to nest in old fields, which were ungrazed and often appeared to be lacking current land-use activities. Our results align with previous work in this geography that located substantial numbers of Mottled Duck nests in permanent pasture characterized by dense perennial grasses and scattered shrubs (Durham and Afton 2003), similar to the old field and pasture designations in this study. We found that emergent marsh was least likely to be selected by Mottled Ducks for nesting. However, we acknowledge that there may have been a positive bias in discovering Mottled Duck nests in upland landscapes, because an individual that consistently logs locations in dry areas indicates a more obvious nesting attempt compared to an individual that logged repeated locations in marsh, which may simply be a loafing location. Perhaps more importantly, ducks that had their nest site in emergent marsh also tended to have most of their 6-day 80% home range in emergent marsh, therefore we were unable to detect differences in used vs. available habitat with our randomly generated points (i.e., one cannot detect preference if only one landscape type is deemed available).

The upland habitats preferred by Mottled Ducks in this study recorded some of the highest nest success rates (> 20%) reported in this region (Bonczek and Ringelman 2021), suggesting that preference for old fields and pasture may be adaptive. However, there was a non-significant trend for nests constructed over water

in emergent marsh to achieve approximately triple the nest success compared to those in upland habitats, albeit sample size was small. Nevertheless, we emphasize the potential importance of emergent marsh to nesting Mottled Ducks, especially because upland habitat in southwest Louisiana is increasingly scarce and is subject to mowing, grazing, and burning during the nesting season.

Interestingly, we found no evidence that some landscape features previously identified as important to Mottled Ducks affected the probability of nest site selection or success. For example, while current management recommendations focus on conserving large tracts of uplands for nesting Mottled Ducks (Hartke 2013), we found no effect of patch size on nest site selection or success. Indeed, we observed that Mottled Ducks used small tracts of habitat successfully, and thus they should not be discounted in conservation measures. Old fields were the most preferred landscape type and averaged only 14 ha, although current management guidelines delineate tracts of < 16 ha as unsuitable habitat (Hartke 2013). Distance to nearest water was another variable that was not competitive in nest site selection analyses, despite the risks associated with overland brood movements and the foraging opportunities provided to the incubating female. We were able to follow five successful females to their brood rearing area. Nest sites were located an average of 2440 ± 1600 m straight-line distance from the nest site to the final brooding area, indicating that broods travel greater distances than previously thought; one female traveled 7.8 km over ~72 hours. This has important management implications: upland nesting areas > 1.6 km from potential brood-rearing areas were deemed unsuitable for Mottled Ducks (Krainyk et al. 2019), but our data indicate such a designation would exclude viable nesting habitat and thus deprioritize areas for conservation that are useful. We observed two Mottled Ducks prospecting brood habitat the day before hatch (see also Casazza et al. 2020) and then leading their brood past seemingly similar wetlands to the final brood-rearing area, highlighting how little is understood about the post-hatch ecology of this species.

At the microhabitat scale, Mottled Ducks preferred nest sites with greater visual obstruction; the relationship was non-linear, with an inflection point at ~6 dm (Fig. 3). An intermediate level of visual obstruction may allow for the earlier detection of predators and thus increased female survival, which may be a trade-off with nest concealment. Nests with greater obstruction may impede the movement of predators or partially obscure olfactory cues (Hines and Mitchell 1983), although the effects of visual obstruction on duck nest survival are inconsistent across studies (reviewed in Ringelman and Skaggs 2019). Here, we found that selection for higher visual obstruction was adaptive and resulted in higher nest survival. The relationship was fairly linear, but the lower confidence interval began to asymptote ~6 dm (Fig. 4). The adaptive preference for denser vegetation at the nest bowl aligns with Durham and Afton (2003) who documented preference for sites with higher visible obstruction ($\mu_{\text{used}} = 7$ dm, $\mu_{\text{random}} = 5$ dm) and differential success based on vegetation density ($\mu_{\text{successful}} = 8$ dm, $\mu_{\text{unsuccessful}} = 7$ dm). Finger et al. (2003) also found Mottled Duck nests where vegetation density > 6 dm, so this preference appears to be widespread across geographies and years. This highlights the importance of providing nesting habitat with vegetation visual obstruction ratings > 6 dm to benefit nesting Mottled Ducks. Human activities such as mowing, grazing, and

burning that reduce vegetation density may therefore limit both the number of Mottled Ducks using that upland habitat, and the nest success of those that do.

This study provides the first defensible estimates of renesting propensity for Mottled Ducks (but see Ringelman et al. 2022 for anecdotal data from geolocators). The renesting rates observed in 2019 and 2020, 0.71 and 0.75, respectively, were higher than those of other closely related dabbling duck species that are typically ~50%, although vary with species and age among other factors (Baldassarre 2014). Mottled Duck renesting propensity declined as the number of days spent incubating the previous clutch increased, similar to other waterfowl species (Krapu et al. 1983, Fondell et al. 2006). Nesting birds must obtain at least a minimum threshold level of nutrient reserves to produce a clutch (Alisauskas and Ankney 1992) and the reduced foraging associated with incubation likely causes a decline in body condition (Hepp et al. 2005), therefore reducing capacity to attempt an additional nest. Initiation date of the first nest was the most important variable influencing renesting propensity in Mallards (*Anas platyrhynchos*) in the Canadian Prairie Parklands (Arnold et al. 2010) and Northern Pintails (*Anas acuta*) on the Yukon-Kuskokwim Delta, Alaska (Flint and Grand 1996). In contrast, initiation date did not influence renesting propensity in Mottled Ducks; this seems logical because Mottled Ducks have a longer nesting window that extends from February through August (Bonczek and Ringelman 2021) and individuals are not as constrained by changes in temperatures or impending autumn migration. The Mottled Duck renesting interval, which averaged 26 days, was much longer than in Mallards, which averaged 5.5 days for females that failed during laying and 10.7 for females that failed during incubation (Arnold et al. 2010). In this study, the two Mottled Ducks that lost their nests during laying did not renest for ≥ 23 days. This suggests either a scarcity of resources to recoup nutrients lost during egg production, less phenological pressure to renest quickly, or both. Renesting did not occur in 2018 when moisture levels were below average but were high in years of average moisture (2019 and 2020). Therefore, although this vital rate has the potential to help compensate for low nest success resulting in a higher hen success rate (Cowardin and Johnson 1979), that may depend on water conditions. Management actions that maintain water on the landscape during years of drought may promote renesting behavior.

Using transmitters helped eliminate some prior biases in habitat selection studies but may have influenced our results in other ways. Equipping ducks with backpack transmitters may have resulted in behavioral and/or physiological changes (Garrettson et al. 2000, Kesler et al. 2014) and reduced nesting propensity (Paquette et al. 1997), but seem unlikely to have altered nest site preferences. The most likely bias associated with our transmitter-based nesting study is the potential to fail to recognize nesting attempts because the nests are depredated before they can be identified by the GPS location data. Given that some of the nests we located failed during the laying period, we probably missed the discovery of other nests depredated during laying; this may be more likely in emergent marsh where repeated locations could be mistaken for loafing. If we missed failed nests, this would bias our nesting propensity and renesting effort estimates low, and our nest success estimates high. We therefore encourage managers to view our numerical estimates in context with other published values where possible (Bonczek and Ringelman 2021).

Given the apparent relationship between old fields, dense vegetation, and Mottled Duck nest site selection and survival, land management practices could be tailored to promote successful Mottled Duck nesting. Currently, most large tracts of upland habitat in southwestern Louisiana are heavily managed and practices such as cattle grazing, haying, and prescribed burns may preclude the production of tall, dense stands of vegetation that are associated with higher probability of use and higher nest survival rates. For example, prescribed burns in fall or early winter are commonly used to reset succession and promote desired plant species, but when burned too late, the vegetation may not rebound in time for the breeding season or nests may be destroyed in spring burns (Nyman and Chabreck 1995). The timing of prescribed burns should allow for sufficient regrowth prior to the nesting season, and burned areas should be rotated on a 3–5 year cycle. More frequent burn cycles may result in an ecological trap if Mottled Ducks are attracted to tall, dense stands of vegetation, only for their nests to be destroyed in a prescribed fire. Most of southwestern Louisiana is privately owned, so to conserve and improve Mottled Duck nesting habitat, it is crucial to educate landowners on management practices that create conditions that will attract nesting females and promote nest success.

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